

FYKE SIMON V. TONEL

Metro Manila, Philippines | 0961-575-4156 | fyketonel22@protonmail.com | fykelabs.vercel.app | github.com/ultraelectronica

Education

STI College Caloocan

Bachelor of Science in Computer Science

Caloocan City, Metro Manila

September 2022 – July 2026

Experience

Back-End Developer (Intern) – Inspire Holdings Inc.

January 2026 – February 2026

- Built and deployed a **NestJS payment backend** on AWS (Lambda, API Gateway, RDS), integrating secure UnionBank APIs.
- Reduced **p95 latency** from **420ms to 180ms (–57%)** and CPU usage by **35%** using Redis caching and token-bucket rate limiting
- Delivered **CI/CD pipelines** and **40+ unit tests** using GitHub Actions and Jest.

Software Developer (Intern) – Socia I.T. Solutions

February 2026 – March 2026

- Integrated backend APIs to support **order processing, transactions, and operational workflows.**
- Developed core modules including **POS system, driver portal, and franchising interface.**
- Implemented **WebSocket-driven real-time features** for live streaming and interactive rewards.

Projects

Flick: Music Player with Custom UAC 2.0

January 2026 – Present

Flutter, Dart, Rust, and Kotlin

- Engineered **four Rust-based audio engines** (standard, high quality, studio-grade, hardware abstraction layer for DAPs) **enabling direct USB output and bypassing Android audio pipeline constraints.**
- Architected a hybrid differential music library scanner using **Android MediaStore, Rust parallel traversal, and fingerprint caching**, reducing scan times from **~11s to 85–328ms (up to 34× faster)** across **1,000+ track SD-card libraries.**
- Open-source project with **2000+ installs/downloads** across **GitHub and Play Store distributions.**

Latch: Secure Private Media Vault Application

November 2025 – Present

Flutter, Dart, and Kotlin

- Architected cryptographically secure storage** using **AES-256-GCM/CTR** dual-engine encryption, **PBKDF2** salted key derivation, and multi-tier authentication (PIN, password, biometrics, Decoy Mode).
- Designed a cross-app pipeline** via platform channels to seamlessly hand off secure audio playback directly to *Flick Player's* **high-performance audio engine.**
- Open-source project with **200+ installs/downloads** across **GitHub and Play Store distributions.**

br41ndmg: High-Fidelity Audio Resampling Library

January 2026– Present

Rust (DSP, Audio Processing)

- Implemented **polyphase sinc-based resampling** for high-accuracy sample rate conversion.
- Achieved **>100 dB stopband attenuation** and **<0.1 dB passband ripple.**

PASADA: Ride-Hailing & Fleet Management Ecosystem.

December 2024 – November 2025

Flutter, TypeScript (Bun/Deno), React, Supabase, and Google Cloud Platforms

- Architected a distributed system spanning **4 applications (Passenger, Driver, Admin, Web)** with shared authentication, RBAC, and a data pipeline for **7-day demand forecasting** and analytics.
- Validated platform scale and UX** through live field testing with **132 active passengers** and **32 assigned modernized jeepney drivers.**

Technical Skills

Languages: Dart, Rust, TypeScript, Kotlin, Java, SQL

Backend: Node.js, NestJS, PostgreSQL, Redis, Supabase, Upstash

Frontend/Mobile: Flutter, React Native, React, Next.js, Vite, HTML, CSS

Cloud/DevOps: AWS, GCP, Docker, Cloudflare, CI/CD, Git, GitHub, GitLab, Codeberg

Systems: WebSockets, API Design, Rate Limiting, Real-Time Systems